

SUJOY P T

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SUMMARY

Data Science postgraduate with expertise in Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and Generative AI. Experienced in developing AI-driven solutions using Python, TensorFlow, PyTorch, Transformers, and RAG architectures. UGC NET qualified with strong analytical, research, and problem-solving skills, seeking opportunities in Data Science and AI/ML.

EDUCATION

University of Kerala, Dept of Futures Studies:- MSc Data Science CGPA: 8.11/10	2023 – 2025
St. Joseph's College (Autonomous), Devagiri:- BSc Mathematics CGPA: 5.97/10	2019 – 2022

ACADEMIC PROJECTS

[GitHub Repositories](#)

Under Water Waste Detection Using YOLOv11

- Developed a deep learning-based marine debris detection system using YOLOv11.
- Performed image preprocessing, augmentation, and model training on underwater datasets.
- Evaluated performance using Precision, Recall, and mAP metrics.
- Technologies: Python, YOLOv11, Roboflow, Computer Vision

AI Tutor Chatbot

- Developed an AI-powered Tutor Chatbot using FastAPI and React with text and voice interaction capabilities.
- Integrated Speech-to-Text (STT) and Text-to-Speech (TTS) modules for seamless conversational experience.
- Implemented Retrieval-Augmented Generation (RAG) to provide accurate and context-aware responses
- Technologies: Python, FastAPI, React, LangChain, LLMs, RAG.

Hate Speech Recognition using BERT

- Fine-tuned a BERT model for automated hate speech and toxic content classification.
- Implemented tokenization, preprocessing, and contextual embedding techniques using Hugging Face Transformers.
- Achieved high classification performance through self-attention-based language understanding.
- Technologies: Python, PyTorch, Transformers, NLP.

Sales Forecasting using Time Series Analysis

- Developed a sales forecasting model using Facebook Prophet on retail sales data.
- Performed preprocessing, outlier handling, and trend analysis.
- Evaluated forecasting performance using RMSE and MAE metrics.
- Technologies: Python, Prophet, Pandas, Matplotlib.

Customer Segmentation using Machine Learning

- Applied K-Means Clustering to segment customers based on spending behavior and income.
- Conducted exploratory data analysis and cluster optimization using the Elbow Method.
- Generated customer insights to support targeted marketing strategies.
- Technologies: Python, Scikit-Learn, Pandas, Seaborn.

TECHNICAL SKILLS

- **Languages:** Python, SQL, R
- **Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Generative AI, RAG
- **Frameworks & Libraries:** Scikit-Learn, TensorFlow, PyTorch, Hugging Face, Matplotlib, Seaborn
- **Tools & Others:** Git, GitHub, Google Colab, Jupyter Notebook, Power BI
- **Databases:** MySQL

SOFT SKILLS

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| • Problem Solving | • Team Collaboration | • Continuous Learning |
| • Analytical Thinking | • Communication | • Research Orientation |

CERTIFICATIONS AND QUALIFICATIONS

- **Data Visualization using Power BI** — *ASAP KERALA*, Issued: July 2024
- **UGC NET Qualified** — *UGC*, Issued: June 2025